

8.7.1] @ Follow Example 8.7-2. $\frac{L(80)}{L(76)} =$

$$\frac{(128\pi)^{-n/2} \exp\left\{-\frac{1}{128} \sum_{i=1}^n (x_i - 80)^2\right\}}{(128\pi)^{-n/2} \exp\left\{-\frac{1}{128} \sum_{i=1}^n (x_i - 76)^2\right\}} = \exp\left\{\left(-\frac{1}{128}\right)\left(-8 \sum_{i=1}^n x_i + n(80^2 - 76^2)\right)\right\}$$

Setting $L(80)/L(76) \leq k$, & taking logarithms, we get

$$\frac{1}{16} \sum_{i=1}^n x_i - n \frac{(80^2 - 76^2)}{128} \leq \ln k, \text{ or } \bar{x} \leq \frac{16 \ln k}{n} + \frac{80^2 - 76^2}{8} =: c.$$

(b) $\alpha = P(\bar{X} \leq c; \mu = 80) = P\left(\frac{\bar{X} - 80}{(8/\sqrt{n})} \leq \frac{c - 80}{(8/\sqrt{n})}; \mu = 80\right)$

$$= P(Z \leq \frac{c - 80}{(8/\sqrt{n})}) = \Phi\left(\frac{c - 80}{8/\sqrt{n}}\right) \approx 0.05.$$

$$\beta = P(\bar{X} > c; \mu = 76) = P\left(\frac{\bar{X} - 76}{8/\sqrt{n}} > \frac{c - 76}{8/\sqrt{n}}; \mu = 76\right) = 1 - \Phi\left(\frac{c - 76}{8/\sqrt{n}}\right) \approx 0.05.$$

Thus $\frac{c - 80}{8/\sqrt{n}} \approx -1.645$ & $\frac{c - 76}{8/\sqrt{n}} \approx 1.645$, or

$$c - 80 \approx \frac{(-1.645)(8)}{\sqrt{n}} \text{ & } c - 76 \approx \frac{(1.645)(8)}{\sqrt{n}}. \text{ This gives}$$

$$80 - \frac{(1.645)(8)}{\sqrt{n}} = 76 + \frac{(1.645)(8)}{\sqrt{n}} \Rightarrow 4 = \frac{(1.645)(16)}{\sqrt{n}} \Rightarrow n \approx 43.3, \text{ &}$$

then substituting this gives $\boxed{c \approx 78.}$ Round n to $\boxed{43}.$

8.7.3] @ The likelihood is $L(\theta) = f(x_1; \theta) \cdots f(x_n; \theta) = \theta^{-n} e^{-\frac{1}{\theta} \sum_{i=1}^n x_i}$, which depends on $(x_1, \dots, x_n)$ only through $x_1 + \dots + x_n$.

(b) $L(3) = 3^{-12} e^{-\frac{1}{3} \sum_{i=1}^{12} x_i}$, $L(5) = 5^{-12} e^{-\frac{1}{5} \sum_{i=1}^{12} x_i}$, $\frac{L(3)}{L(5)} = \left(\frac{5}{3}\right)^{12} \exp\left(-\frac{2}{15} \sum_{i=1}^{12} x_i\right) \leq k$; this is equivalent to $\sum_{i=1}^{12} x_i \geq c$ for some c .

$$\alpha = P\left(\sum_{i=1}^{12} X_i \geq c; \theta = 3\right) = P\left(\frac{2}{\theta} \sum_{i=1}^{12} X_i \geq \frac{2c}{\theta}; \theta = 3\right) = P\left(\frac{2}{3} \sum_{i=1}^{12} X_i \geq \frac{2}{3}c; \theta = 3\right) = P(\chi^2(24) \geq \frac{2}{3}c) = 0.1 \Rightarrow \frac{2}{3}c = \chi_{0.1}^2(24) = 33.2 \Rightarrow c = 49.8.$$

Thus critical region is $\{\sum_{i=1}^n x_i \geq 48.3\}$ or $\{\bar{x} \geq 4.025\}$.

⑥ The critical region in ⑥ only depended on the fact that $H_0: \theta = 3$ & $H_1: \theta = \theta_1$, where $\theta_1 > 3$, so it will be the same here: $\{\bar{x} \geq 4.025\}$.

⑦ As in Example 8.7-4, the answer is yes: we get the same region no matter what simple alternative $H_1: \theta = \theta_1$, $\theta_1 > 3$ is given.

8.7.6) a. For c_1 : $P(\bar{X} \geq c_1; \mu = 80) = P\left(\frac{\bar{X} - 80}{3/\sqrt{n}} \geq \frac{c_1 - 80}{3/\sqrt{n}}; \mu = 80\right)$
 $= P(Z \geq \frac{c_1 - 80}{3/4}) = 0.05$, i.e. $\frac{c_1 - 80}{3/4} = 1.645$ or $c_1 = 81.23$.

For c_2 : $P(\bar{X} \leq c_2; \mu = 80) = P(Z \leq \frac{c_2 - 80}{3/4}) = 0.05 \Rightarrow$
 $\frac{c_2 - 80}{3/4} = -1.645 \Rightarrow c_2 = 78.77$. (Could also find from c_1 , by symmetry.)

For c_3 : $P(|\bar{X} - 80| \geq c_3; \mu = 80) = P\left(|\frac{\bar{X} - 80}{3/4}| \geq \frac{c_3}{3/4}; \mu = 80\right)$
 $= P(|Z| \geq \frac{4}{3}c_3) = 0.05 \Rightarrow \frac{4}{3}c_3 = 1.96 \Rightarrow c_3 = 1.47$.

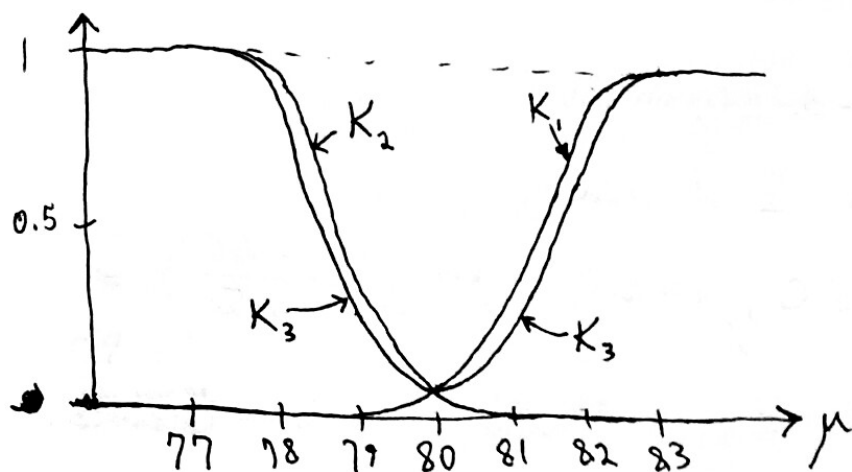
⑥ $K_1(\mu) = P(\bar{X} \geq 81.23; \mu) = P\left(\frac{\bar{X} - \mu}{3/4} \geq \frac{81.23 - \mu}{3/4}; \mu\right)$
 $= P(Z \geq \frac{4}{3}(81.23 - \mu)) = 1 - \Phi\left(\frac{4}{3}(81.23 - \mu)\right)$.

$K_2(\mu) = P(\bar{X} \leq 78.77; \mu) = P\left(Z \leq \frac{4}{3}(78.77 - \mu)\right) = \Phi\left(\frac{4}{3}(78.77 - \mu)\right)$

$K_3(\mu) = P(|\bar{X} - 80| \geq 1.47; \mu) = \dots$

$= P(\bar{X} \geq 81.47; \mu) + P(\bar{X} \leq 78.53; \mu) = P\left(Z \geq \frac{81.47 - \mu}{3/4}\right) +$

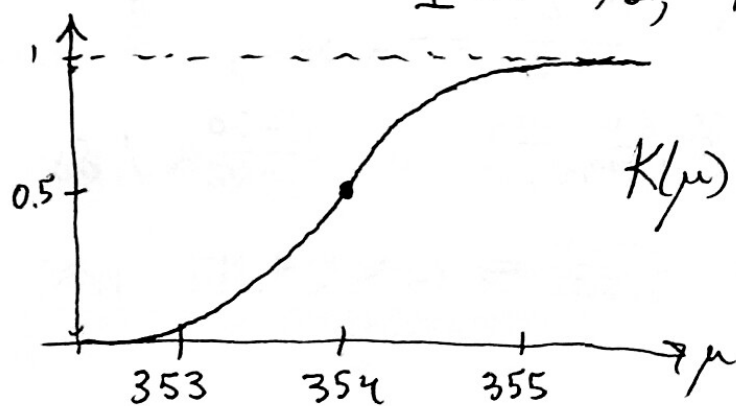
$P\left(Z \leq \frac{78.53 - \mu}{3/4}\right) = \Phi\left(\frac{4}{3}(78.53 - \mu)\right) + 1 - \Phi\left(\frac{4}{3}(81.47 - \mu)\right)$.



8.6-2 a) $K(\mu) = P(\bar{X} \leq 354.05; \mu) = P\left(\frac{\bar{X} - \mu}{2/\sqrt{12}} \leq \frac{354.05 - \mu}{2/\sqrt{12}}\right)$
 $= P(Z \leq \frac{354.05 - \mu}{2/\sqrt{12}}) = \Phi\left(\frac{354.05 - \mu}{2/\sqrt{12}}\right).$

b) $\alpha = K(355) = \Phi\left(\frac{354.05 - 355}{2/\sqrt{12}}\right) = \Phi(-1.645) \approx \boxed{0.05}.$

c) $K(354.05) = \Phi(0) = 1/2$, $K(353.1) = \Phi\left(\frac{354.05 - 353.1}{2/\sqrt{12}}\right) = \Phi(1.645) \approx 0.95.$



d) $\bar{X} = 353.83$. Since $\bar{X} \leq 354.05$, we reject H_0 .

e) $P(\bar{X} \leq 353.83; \mu = 355) = P\left(Z \leq \frac{353.83 - 355}{2/\sqrt{12}}\right)$
 $= \Phi(-2.026) \approx \boxed{0.0212}.$

1. a) Type I error $= P(\bar{X} \leq 57; \mu = 60) = P\left(Z \leq \frac{57 - 60}{10/\sqrt{20}}\right)$
 $= \Phi(-1.342) \approx 0.09$. Type II error $= P(\bar{X} > 57; \mu = 50)$

$$= P(Z > \frac{57-50}{10/\sqrt{20}}) = \Phi(-3.130) \approx 0.$$

(b) Type I ~~decreases~~, Type II ~~decreases~~ ^{increases}.

(c) Want $0.01 = P(\bar{X} \leq c; \mu=60) = P(Z \leq \frac{c-60}{10/\sqrt{20}}) = \Phi(\frac{c-60}{10/\sqrt{20}})$, i.e. $\frac{c-60}{10/\sqrt{20}} \approx -Z_{0.01} = -2.326$, i.e. $c \approx 54.80$.

So critical region is $\{\bar{X} \leq 54.80\}$. Type II error =

$$P(\bar{X} > 54.80; \mu=50) = P(Z > \frac{54.80-50}{10/\sqrt{20}}) = \Phi(2.147) \approx 0.016.$$

~~0.016~~ $\approx 1 - \Phi(2.147) \approx 0.016$.

(d) Want c, n so that $0.01 = P(\bar{X} \leq c; \mu=60) =$

$$P(Z \leq \frac{c-60}{10/\sqrt{n}}) = \Phi(\frac{c-60}{10/\sqrt{n}}), \text{ i.e. } \frac{c-60}{10/\sqrt{n}} \approx -2.326; \text{ \& } 0.03 = P(\bar{X} > c; \mu=50) = 1 - \Phi(\frac{c-50}{10/\sqrt{n}}), \text{ i.e. } \frac{c-50}{10/\sqrt{n}} \approx 1.88.$$

Solving gives $\frac{c-60}{-2.326} = \frac{c-50}{1.88} \Rightarrow c \approx 54.47, n \approx$

$17.7 \approx \boxed{18}$.

[2] $\bar{X} - \bar{Y} \sim N(0, \frac{400+225}{n})$. $K(0) = P(\bar{X} - \bar{Y} \geq c; 0) =$

$$P(Z \geq \frac{c-0}{25/\sqrt{n}}) = 1 - \Phi(\frac{c-0}{25/\sqrt{n}}). \text{ \& } 0.05 = K(0) = 1 - \Phi(\frac{c}{25/\sqrt{n}})$$

$$\Rightarrow \frac{c}{25/\sqrt{n}} = 1.645. \text{ \& } 0.9 = K(10) = 1 - \Phi(\frac{c-10}{25/\sqrt{n}}) \Rightarrow \frac{c-10}{25/\sqrt{n}} = -1.282.$$

Solving gives $n \approx 53.5 \nearrow \boxed{54}$ and $c \approx \boxed{5.594}$.